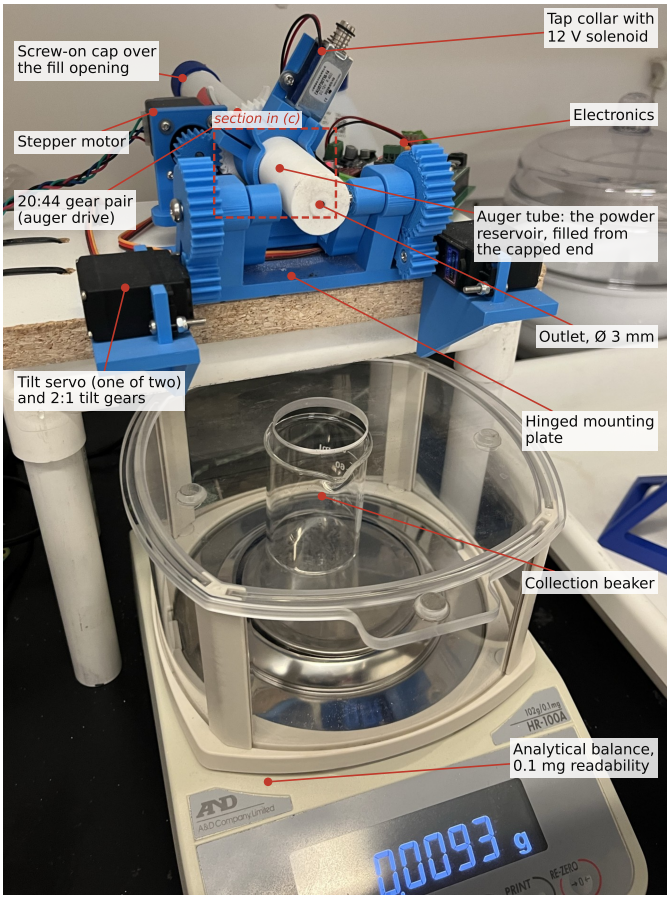
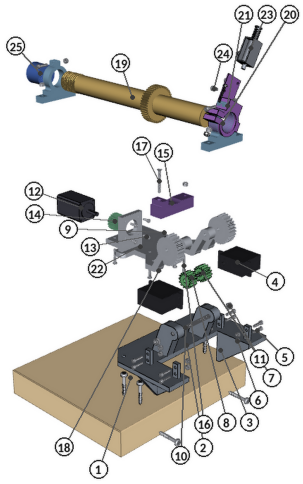
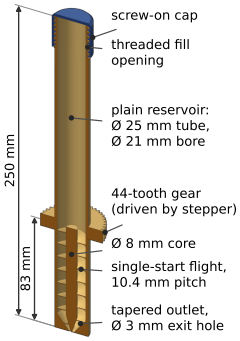


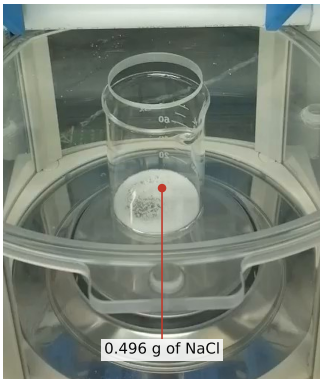
(a) As tested



(b) Exploded view

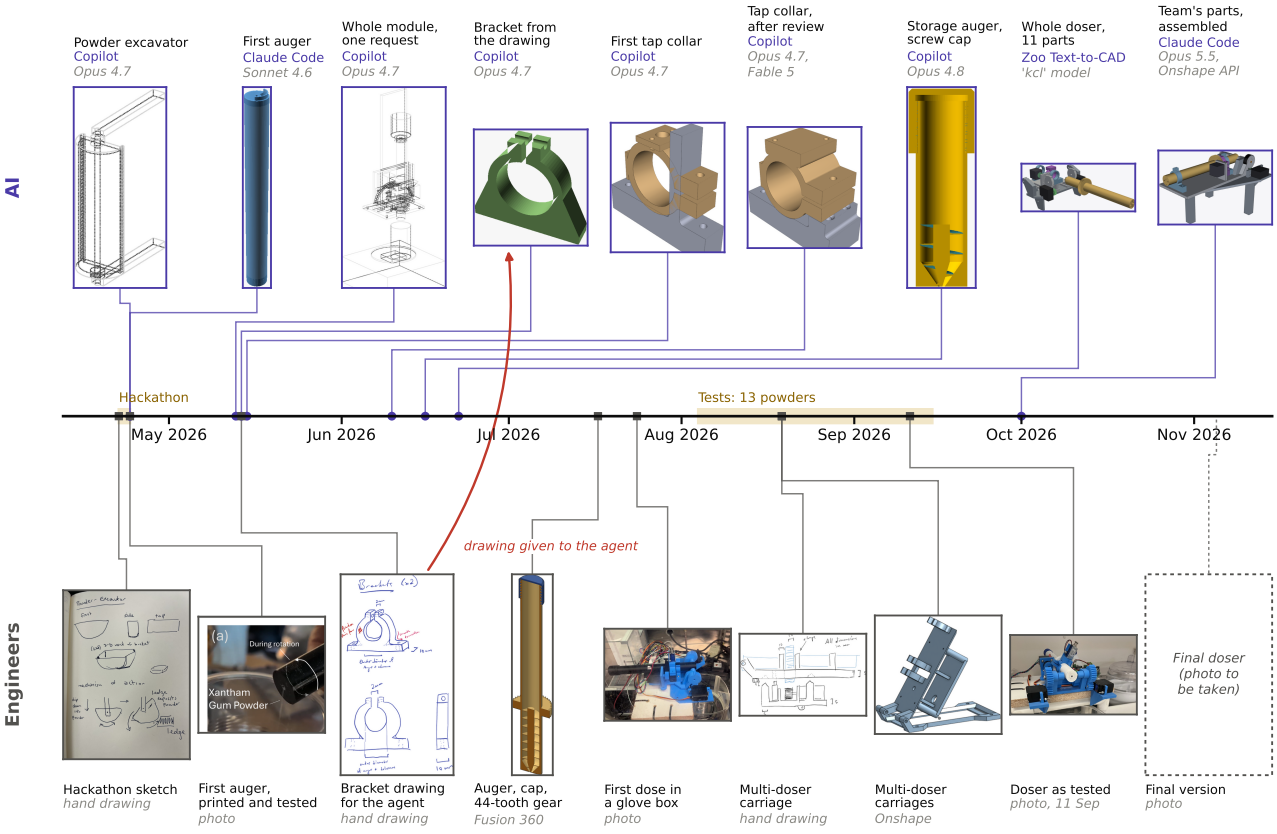


(c) Auger section

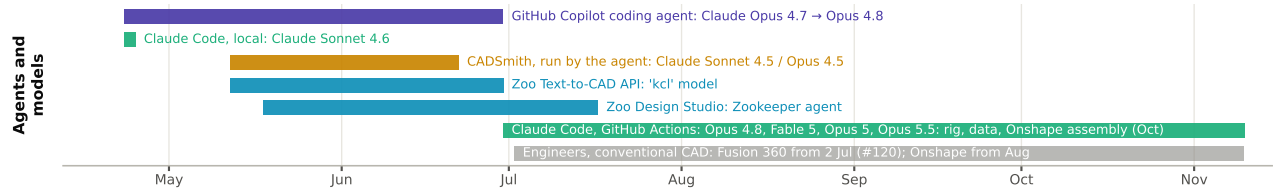


(d) Fixed camera

Figure 1: The doser. One printed auger tube is both the powder reservoir and the dosing screw; the doser turns, tilts and taps it over a cup on an analytical balance. (a) The doser as tested in the second round (photographed 11 Sep 2026), with its parts labelled; the dashed box marks the part drawn in section in (c). (b) Exploded view of the current design, assembled in CAD from the team's parts (#170); the numbers refer to the bill of materials in the SI. (c) Section through the tested auger and its screw-on cap, the team's Fusion 360 part: the flight fills only the outlet end, and the 44-tooth gear is on the outside of the tube. (d) The fixed livestream camera's view of the cup after a 0.5 g NaCl dose (15 Sep 2026; 0.496 g delivered; #116).

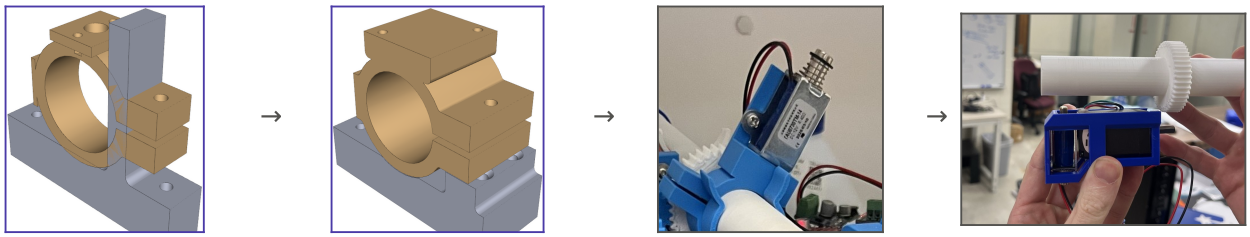


(a) Designs over time

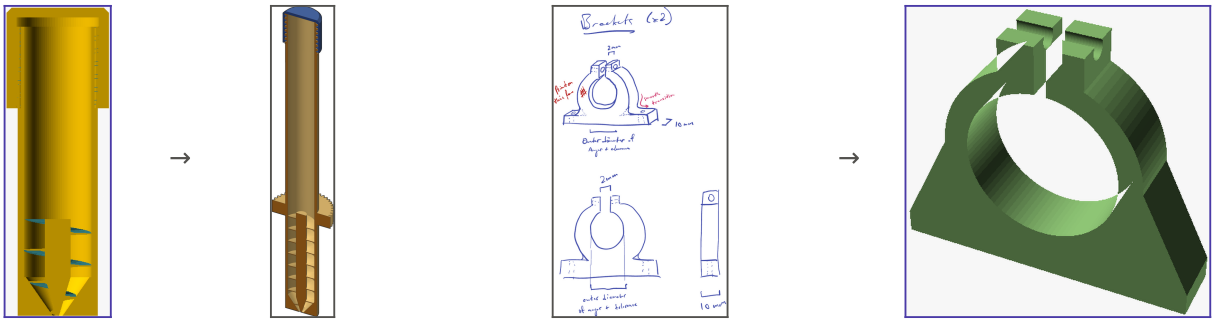


(b) Agents and models

Figure 2: AI agents and engineers, April to November 2026. AI agents modelled the first parts from text and drawings; from July the engineers redrew in conventional CAD the parts that were built and tested, and the agents moved on to running the tests and assembling the CAD. (a) Key designs on one date axis, each joined to its date. Above the axis are the designs modelled by AI agents, with the agent (purple) and model (grey) under each: the powder excavator (#2) and the first auger (#16), from the hackathon; the whole module from a single request (#35); the bracket the agent modelled from the team's dimensioned drawing (#46, #47; red arrow); the first tap collar and the collar after review (#51); the storage auger with its screw cap (#49); all 11 parts from one prompt through the Zoo Text-to-CAD API (#104, #110); and the team's parts assembled through the Onshape API (#170). Below the axis is the engineers' work: the hackathon sketch (#2), the first printed auger, the bracket drawing (#46), the Fusion 360 auger, cap and 44-tooth gear that turned in every test (#120), the first dose in a glove box (#117), the multi-doser carriage sketch and Onshape model (#128), and the doser as tested (#156). The hackathon (22–24 Apr) and the 13-powder test campaign (4 Aug–15 Sep) are shaded on the axis. (b) When each agent and model was in use, on the same date axis.



(a) Tap collar



(b) Auger **(c)** Bracket

Figure 3: The original AI parts and what replaced them. Every AI-modelled part was reviewed, printed and tested, but the tested doser ran on later versions: the agent revised the tap collar and the team then rebuilt it, and the team redrew the agent's auger in Fusion 360. AI parts have purple frames, the team's grey. (a) Tap collar: the first AI collar (15 May) and the collar after review (#51, by 10 Jun), the collar on the rig during the tests (11 Sep, #156), and the team's new block (8 Oct, #128). (b) Auger: the AI storage auger with 16:48 gears (16 Jun, #49) and the team's Fusion 360 recreation with 20:44 gears (17 Jul, #120), which turned in every test. (c) Bracket, the other direction: the team's dimensioned drawing (#46) and the agent's model of it (#47), both 14 May.

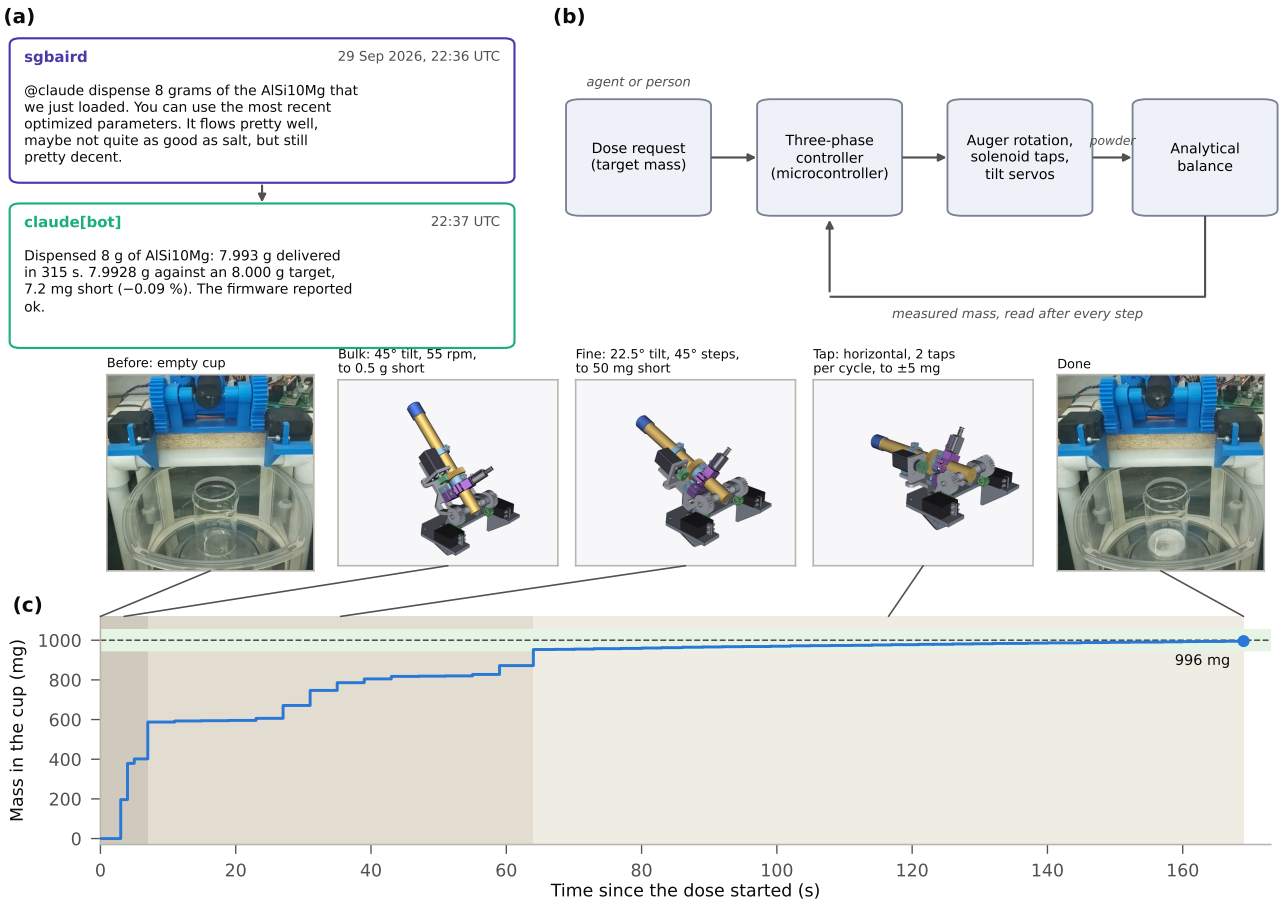
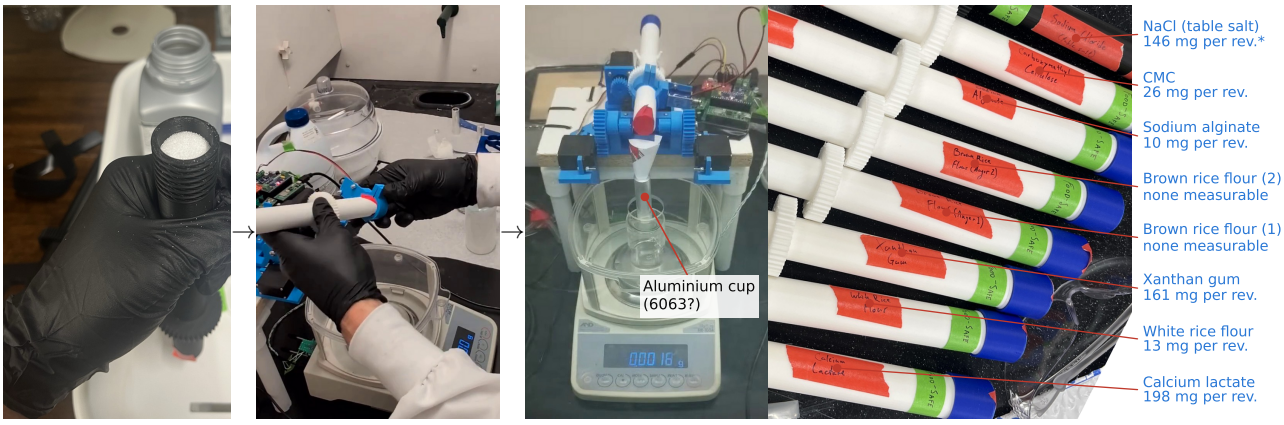
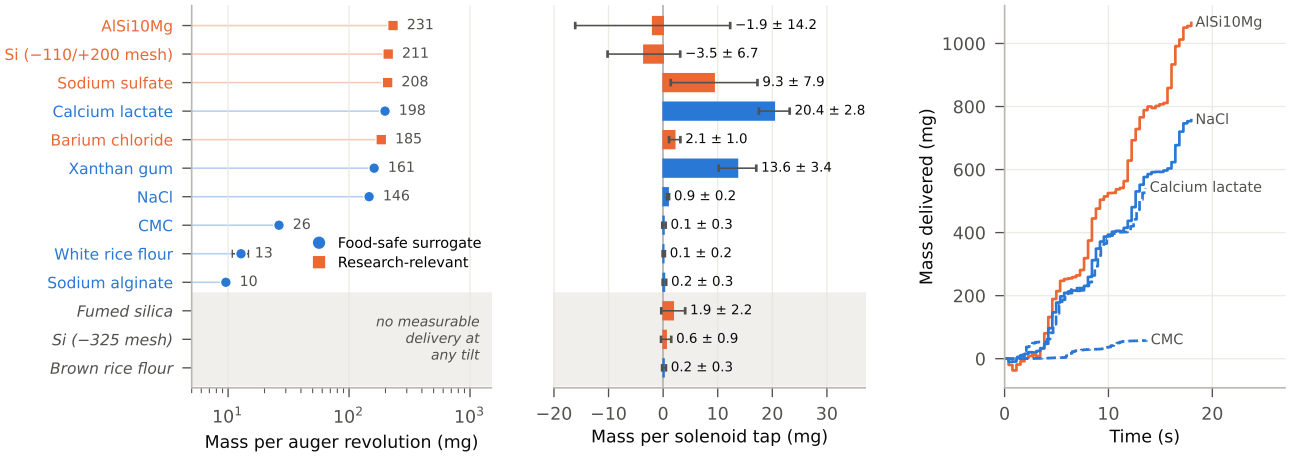


Figure 4: How a dose runs. A person asks the AI agent for a dose in a GitHub comment; a three-phase controller on the doser's microcontroller then turns, tilts and taps the auger until the balance reads the target. (a) A request and the agent's reply (8 g of AlSi10Mg for an atomizer charge, #166, 29 Sep 2026). (b) The control loop: the balance is read after every step, and the controller acts on the mass still to go. (c) One 1 g NaCl dose from the test campaign (12 Aug 2026), with its three phases shaded: bulk, fine and tap. Above the trace are the tube positions in each phase (CAD renders of the current design at 45°, 22.5° and 0°) and the fixed camera's stills before and after a 0.5 g salt dose (15 Sep 2026). The green band is the ±5% acceptance limit. One image rather than subfigures: the lines join the stills to the trace.



(a) Fill **(b)** Fit **(c)** Over the cup **(d)** One auger per powder

Figure 5: Using the doser. Each powder has its own auger, filled once and stored loaded, so changing powder means swapping a tube. (a) Filling an auger with salt through its open end, then capping it (29 Jul 2026). (b) Fitting a filled auger to the doser, and (c) the doser ready over the cup, frames from the 30 Sep 2026 dose video: the powder falls through a paper funnel into an aluminium cup that stands in the glass beaker. (d) Eight of the labelled augers, with each label and the mass one auger revolution delivered (*at 22.5° tilt; Fig. 6a).



(a) Per revolution **(b)** Per tap **(c)** In time

Figure 6: Open-loop delivery of 13 powders under one fixed set of settings. Ten powders moved, from 10 to 231 mg per auger revolution, and three delivered nothing measurable; one solenoid tap released up to 20 mg, but for the fastest powders a single tap was unpredictable. (a) Mass per auger revolution at 22.5° (protocol C; mean ± standard error of six revolutions; log axis). (b) Mass released by one solenoid tap at 22.5° (protocol E): mean ± standard deviation of eight taps, the spread to expect from a single tap. Table S1 gives the standard errors. (c) Mass against time while the auger turned continuously at 15 rpm (protocol D).

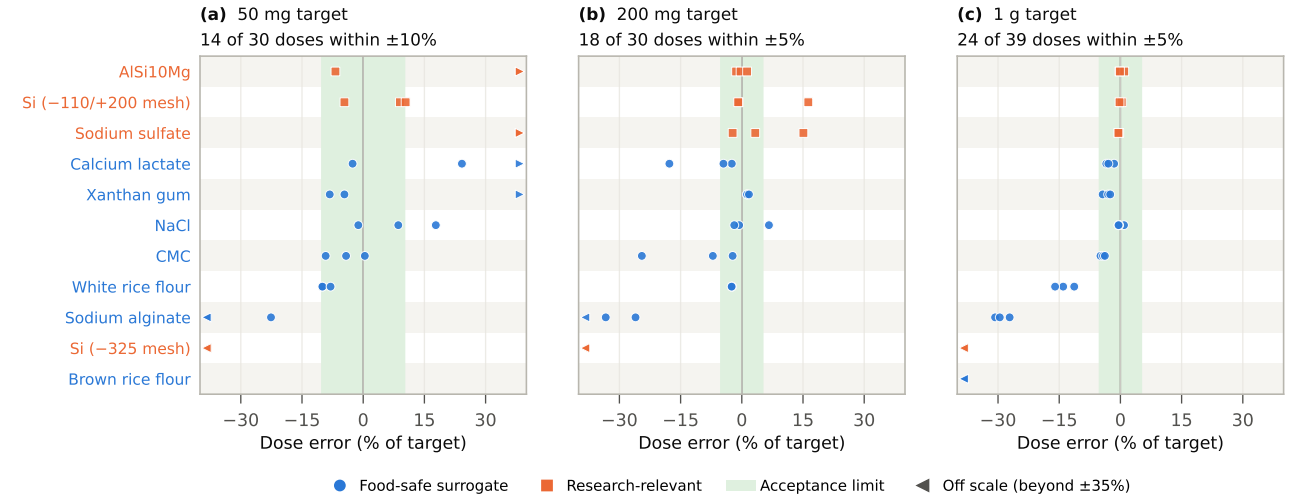


Figure 7: Closed-loop doses. Doses of 1 g landed within ±5% for every powder that delivered at least 26 mg per revolution; smaller doses missed more often, fast powders tending to overshoot and slow ones to fall short. Error of every valid closed-loop dose from both test rounds, one row per powder, ordered by mass per revolution (fastest at the top). The green band is the acceptance limit: ±10% below 100 mg, ±5% at and above. Times per dose are in Table 3 and SI Fig. S3. One image rather than subfigures: the three panels share the powder axis.

Table 1: The 13 powders. Mass per auger revolution: protocol C mean at 22.5° (Fig. 6a). Closed-loop columns: valid doses that finished within the acceptance limit (±10% at 50 mg, ±5% at 200 mg and 1 g) out of the number run (Fig. 7); a dash means not run. The silicon grades are named by sieve size: −325 mesh passes a 44 μm sieve, and −110/+200 mesh lies between about 75 and 150 μm.

Powder	Group	mg per rev.	50 mg	200 mg	1 g	Observed behaviour
AlSi10Mg (gas-atomized)	Research	231	1/3	3/3	3/3	Fastest powder; fluid-like; overshoots small targets
Silicon, −110/+200 mesh	Research	211	2/3	2/3	3/3	Free-flowing
Sodium sulfate	Research	208	0/3	2/3	3/3	Free-flowing; overshoots small targets
Calcium lactate	Surrogate	198	1/3	2/3	3/3	Free-flowing; largest tap increment
Barium chloride	Research	185	–	–	–	Absorbed moisture and caked in the tube in round 2
Xanthan gum	Surrogate	161	2/3	3/3	3/3	Flows under rotation; can clog after bursts
Sodium chloride (NaCl)	Surrogate	146	2/3	2/3	6/6	Free-flowing baseline (control)
Carboxymethyl cellulose (CMC)	Surrogate	26	3/3	1/3	3/3	Slow; charges electrostatically
White rice flour	Surrogate	13	3/3	3/3	0/3	Slow and pulsing; most variable between revolutions
Sodium alginate	Surrogate	10	0/3	0/3	0/3	Slowest powder that moved
Brown rice flour	Surrogate	none	–	–	0/3	No delivery at any tilt
Silicon, −325 mesh	Research	none	0/3	0/3	0/6	No delivery at 0 or 22.5°; one pulse at 45°
Fumed silica	Research	none	–	–	–	No delivery; very light and fluffy

Table 2: Test protocols. Every powder went through the same protocols in the same order with frozen settings, so results can be compared across powders. The sequence characterizes the powders and does not tune anything. Rotations and speeds refer to the auger, and tilts are the tube's angle above horizontal, from 0° (horizontal park) to 45° (the steepest used). A trial is one recorded measurement. SI Table S**??** gives the settings as the raw records store them and the number of records each protocol produced.

Protocol	Varied	Fixed settings	What it measures	Trials
A Balance baseline	Nothing	Tilt 22.5°; eight readings, no actuation	Noise floor of the run, against which every other protocol is read	8
B Static hold	Tilt (0, 22.5, 45°)	15 s hold at each tilt, no actuation	Whether any powder flows on its own (clean shut-off)	3
C Rotation	Tilt (0, 22.5, 45°)	Six single revolutions per tilt at 30 rpm, with a settled reading after each	Mass per revolution and its revolution-to-revolution variation	18
D Speed sweep	Speed (15, 45, 90 rpm)	Tilt 22.5°; three revolutions commanded per speed; balance streamed every 0.25 s	Delivery rate against speed; mass-against-time traces	3
E Tap	Tilt (0, 22.5°)	Eight trials of a one-revolution re-feed followed by one 60 ms solenoid tap	Mass released by a single tap, with the re-feed logged separately	32
F Vibration	Tilt (0, 22.5°)	As E, with three vibration bursts in place of the tap	Mass released by vibration (no data: driver fault)	0
G Closed-loop, 1 g	Nothing	Three doses of 1.000 g; three-phase controller	Dose error, time, and effort per phase	3 doses
H Closed-loop, small	Target (50, 200 mg)	Three doses at each target; same controller	Dose error and time at small targets	6 doses

Table 3: Closed-loop doses by target mass (protocols G and H, both rounds). Times are given here rather than as a figure panel; SI Fig. SS3 shows every dose.

Target	Limit	Within limit	Median time (s)	Misses that overshot
50 mg	±10%	14 of 30	52	10 of 16
200 mg	±5%	18 of 30	152	3 of 12
1 g	±5%	24 of 39	185	0 of 15

Table 4: The AI tools used, what each did, when, and with which models. Dates and models are taken from the workflow runs, pull requests and commit trailers listed in `paper/figures/data/ai_tools_timeline.csv`; Fig. 2 places the key designs on the same time axis.

Tool	What it did here	When	Models
GitHub Copilot coding agent	Modelled the AI-made parts as CadQuery or OpenSCAD code, one pull request per part; rendered views and ran solid and interference checks	23 Apr – 30 Jun 2026 (CAD)	Claude Opus 4.7, then Opus 4.8
Claude Code, run locally	First Archimedes auger (OpenSCAD) at the hackathon	23–24 Apr 2026	Claude Sonnet 4.6
CADSmith (multi-agent text-to-CadQuery), run by the coding agent	Mounting plate and baseplate; whole doser from the issue #104 prompt	May – Jun 2026	Claude Sonnet 4.5; Opus 4.5
Zoo Text-to-CAD API	KCL parts from text; whole doser (11 parts) from the issue #104 prompt	May – Jun 2026	Zoo 'kcl' model
Zoo Design Studio, Zookeeper agent	Conversational modelling in a live CAD view: the tap collar (three iterations), a servo pinion, multi-doser parts	May – Jul 2026	Zookeeper (model not recorded)
Claude Code, GitHub Actions	Ran the test campaign on the rig, firmware, data analysis, figures; assembled the team's parts through the Onshape API (Oct)	30 Jun – Nov 2026	Claude Opus 4.8, Fable 5, Opus 5, Opus 5.5
Edison Scientific	Literature searches and analysis-mode reviews (including mock peer review)	Apr – Oct 2026	job types only (PaperQA3)
Engineers, conventional CAD (not AI)	Recreated the parts that were tested (Fusion 360, #120); multi-doser (Onshape)	From 2 Jul 2026	–

Table 5: Open-loop delivery for every powder (protocols C and E, round 1, one run per powder). Mass per auger revolution (mg, mean ± standard error of six revolutions) with the tube horizontal (0°) and tilted to 22.5° and 45°; the revolution-to-revolution coefficient of variation (CV) at 22.5°; and the mass released by one solenoid tap at 22.5° (mean ± standard error of eight taps). n.r.: not resolved, meaning the mean did not exceed twice its standard error. Tap values in italics are within the balance noise of their run; fumed silica's only run failed screening. Replaces the tilt panel of the earlier draft (Fig. 4a).

Powder	Group	0°	22.5°	45°	CV at 22.5° (%)	Per tap (mg)
AlSi10Mg (gas-atomized)	Research	49 ± 7	231 ± 13	339 ± 6	13	*-1.9 ± 5.0*
Silicon, −110/+200 mesh	Research	57 ± 7	211 ± 6	302 ± 5	7	*-3.5 ± 2.4*
Sodium sulfate	Research	51 ± 8	208 ± 13	244 ± 12	15	9.3 ± 2.8
Calcium lactate	Surrogate	47 ± 5	198 ± 2	232 ± 3	2	20.4 ± 1.0
Barium chloride	Research	23 ± 4	185 ± 13	200 ± 31	17	2.1 ± 0.4
Xanthan gum	Surrogate	24 ± 2	161 ± 8	187 ± 3	12	13.6 ± 1.2
Sodium chloride (NaCl)	Surrogate	38 ± 4	146 ± 11	265 ± 13	18	0.9 ± 0.1
Carboxymethyl cellulose (CMC)	Surrogate	2.6 ± 0.5	26 ± 1	n.r.	9	*0.1 ± 0.1*
White rice flour	Surrogate	3.8 ± 0.3	13 ± 2	37 ± 7	37	*0.1 ± 0.1*
Sodium alginate	Surrogate	0.8 ± 0.2	9.6 ± 0.3	11 ± 1	8	0.2 ± 0.1
Brown rice flour	Surrogate	n.r.	n.r.	n.r.	–	*0.2 ± 0.1*
Silicon, −325 mesh	Research	n.r.	n.r.	n.r.	–	*0.6 ± 0.3*
Fumed silica	Research	n.r.	n.r.	n.r.	–	*1.9 ± 0.8*†

Table 6: Failure modes of the AI modelling tools observed in this project, and what caught or prevented each one.

Failure	Example	Caught or prevented by
Parts that do not fit together	Whole-module prompts; first tap collar; bracket with unconnected plates (Fig. **??**a,d)	Asking for one part at a time, with interfaces listed
Unrequested changes	Fixing one defect altered an unrelated feature	Full re-inspection of rendered views at every iteration
Mislabelled variants	Two of four test nozzles swapped relative to their documentation	Human cross-check before printing
Sub-millimetre errors	Height mismatch of one print layer	Only the failed print
Designing around bad inputs	Out-of-date neighbouring parts led to a hole, platforms, then a gap (Fig. **??**e–h)	Supplying the correct upstream files
No sense of "finished"	Late iterations disturbed features that were already right	The team freezing the geometry
Wrong orientation	Solenoid mis-oriented on the tap collar over several coding-agent iterations	Switching the part to Zoo Design Studio

One prompt, three tools: the whole doser (issue #104 prompt, 22 Jun 2026)

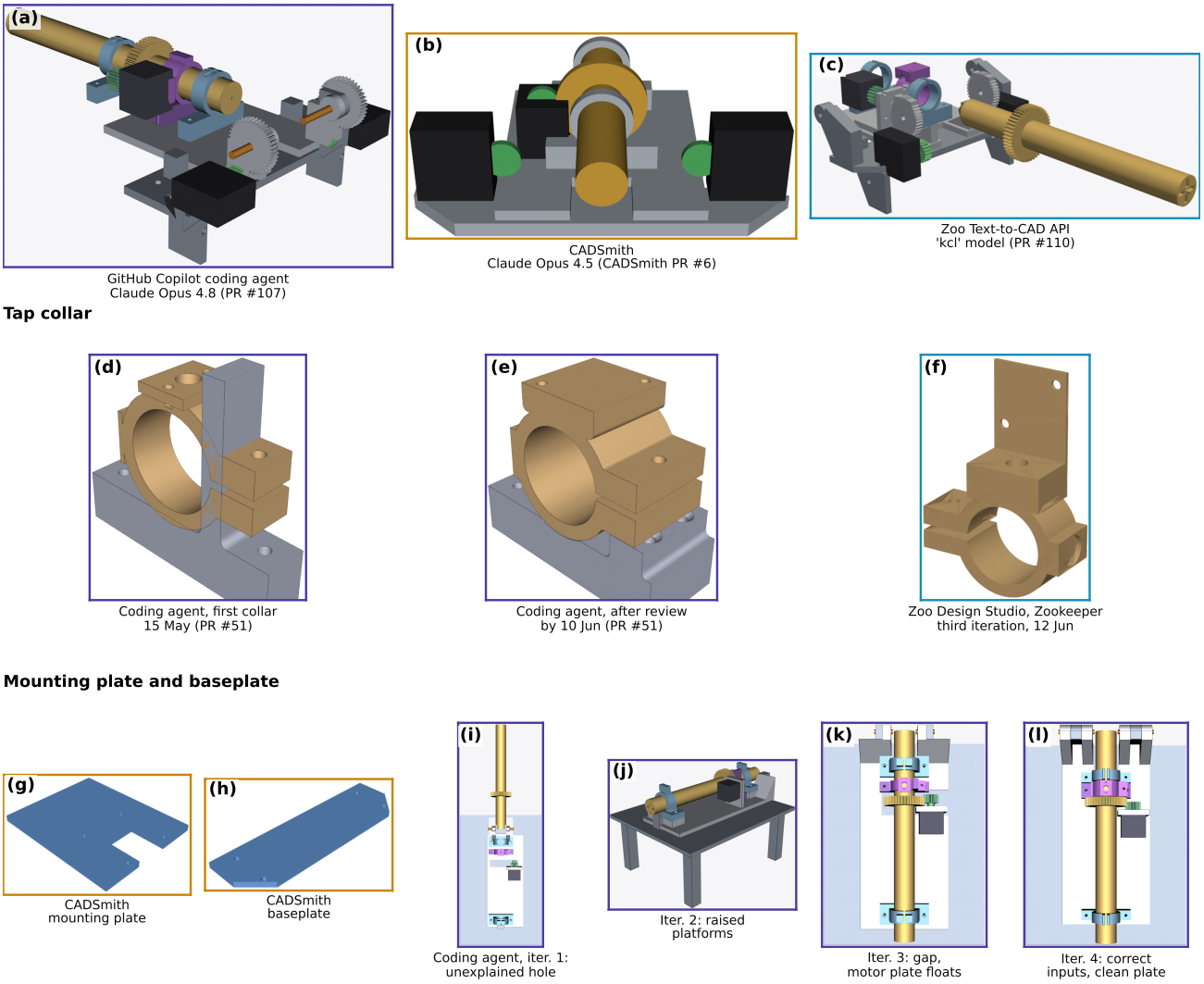


Figure S1: Generative-CAD comparison. Asked for the whole doser at once, every tool returned parts that did not make a working assembly; one part at a time, with review, worked better. (a–c) The same prompt (#104, 22 Jun 2026) given to the GitHub Copilot coding agent, CADSmith and the Zoo Text-to-CAD API, each rendered from the same viewpoint. (d–f) The tap collar from the coding agent (first version and after review) and from Zoo Design Studio (third Zookeeper iteration). (g, h) CADSmith's mounting plate and baseplate. (i–l) The coding agent's four mounting-plate iterations: out-of-date input files led to a hole, then platforms, then a gap, until the correct files were supplied. Not yet reviewed; still one image.

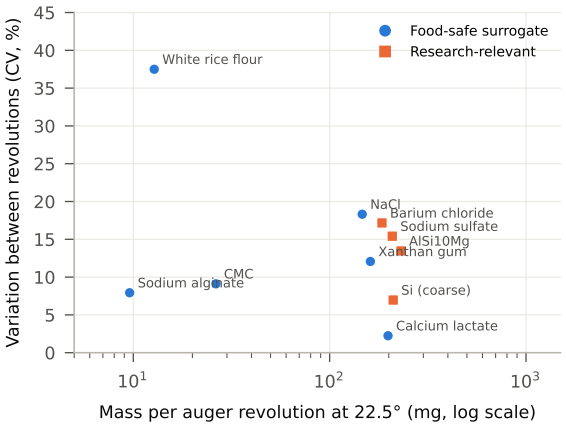


Figure S2: Variation between revolutions. Most powders varied by 2–18% from one revolution to the next; white rice flour, at 37%, varied most. Revolution-to-revolution coefficient of variation against mass per revolution at 22.5° (old Fig. 3c, without the slow/fast shading). The same numbers are in Table S1, so this figure can also be dropped.

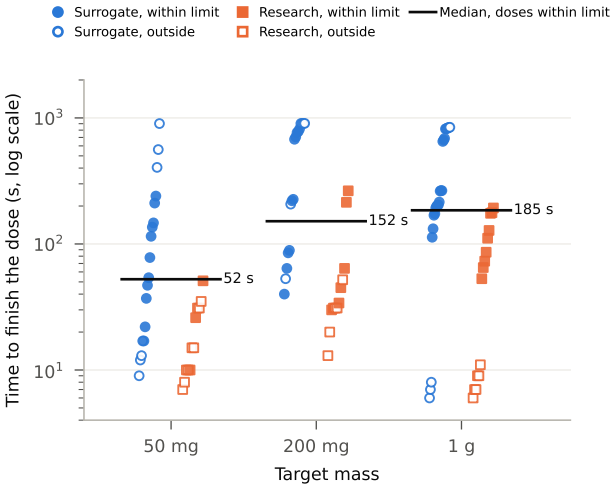


Figure S3: Time per dose. Larger doses took longer: for doses within the limit, the median was 52 s at 50 mg, 152 s at 200 mg and 185 s at 1 g. Time to finish each dose, by target mass; lines are the medians of the doses that finished within the acceptance limit (old Fig. 5d).